

Importance of N_2H_x Reaction Pathways in Modelling of $\text{NH}_3/\text{H}_2/\text{Air}$ Flame Chemistry

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- Develop a reduced kinetic mechanism optimized for NH_3 , NH_3/CH_4 and NH_3/H_2
 - Prioritizing unstretched laminar burning velocity predictions
 - Viable for cases with varying pressure and equivalence ratio
- Validate the Present Mechanism with experimental flame speed measurements from literature

- Unstretched laminar burning velocity, S_L , of premixed flames calculated with 1D unstretched adiabatic simulations
 - Premixed Laminar Flame Speed Calculation model in ANSYS Chemkin-PRO used
 - Computational domain of 10 cm
 - Adaptive grid settings GRAD and CURV set to 0.01
- X_{H_2} , the ratio of H_2 in the fuel, was varied
 - Pressure was kept constant at 1 atm
 - Fuel-oxidizer ration maintained at stoichiometric equivalence ratio

